



Project :

CREDIT CARD FRAUD TRANSACTIONS

SQL QUERIES





PROJECT OVERVIEW

CREDIT CARD FRAUD ANALYSIS



MAIN OBJECTIVES

- Identify and analyse Fraudulent Transactions
- Analyse Fraud Patterns across Merchant Categories
- Evaluate Fraud Rate by Authentication Methods
- Classify Fraud Risk based on Merchant Risk Score
- Identify High-Value Fraudulent Transactions
- Build reusable SQL Objects for Fraud Analysis



SQL CONCEPTS USED

- Views
- JOINS
- Common Table Expressions (CTEs)
- Window Functions
- Aggregations
- CASE Expressions
- GROUP BY & HAVING
- Stored Procedures (Parameterized)



KEY INSIGHTS

- **1.70 %**
Overall Fraud Rate across all Transactions
- **4.99 %**
Highest Fraud Rate in **Financial** Category
- **4.44 %**
Highest Fraud Rate with **No Authentication**
- **\$442.41**
Highest Average Transaction Amount in **Travel** Category
- **\$3356.46**
Highest single Fraudulent Transaction in **Gaming** Category



TASK 1.

FRAUD TRANSACTIONS VIEW

```
CREATE VIEW Fraud_Transaction_View as
SELECT Transaction_ID, Transaction_Amt_USD,
       Merchant_Category, Card_Type, Authentication_Method,
       Transaction_Mode, Customer_Age, Customer_Age_Group,
       Merchant_Risk_Score, Merchant_Risk_Level, Fraud_Status
FROM Credit_Card_Fraud
WHERE Fraud_Status = 1;
GO

SELECT * FROM Fraud_Transaction_View
GO
```

	Transaction_ID	Transaction_Amt_USD	Merchant_Category	Card_Type	Authentication_Method	Transaction_Mode	Customer_Age	Customer_Age_Group	Merchant_Risk_Score	Merchant_Risk_Level	Fraud_Status
1	64	2396.82	Crypto Exchange	Visa	OTP	ATM	23	18-25	92	Very High	1
2	149	18.34	Online Retail	Visa	3D Secure	Online	81	61+	18.6	Very Low	1
3	268	7.25	Gaming	Visa	No Authentication	ATM	28	26-35	67.1	High	1
4	364	246.9	Crypto Exchange	Amex	OTP	POS	34	26-35	85.7	Very High	1
5	392	576.76	Crypto Exchange	Visa	OTP	Online	72	61+	78.4	High	1
6	397	421.53	Groceries	Visa	No Authentication	Online	53	46-60	34.6	Low	1
7	404	92.37	Electronics	Visa	PIN	ATM	75	61+	80.1	Very High	1
8	600	23	Gift Cards	Visa	No Authentication	Online	58	46-60	100	Very High	1
9	798	28.26	Online Retail	Amex	PIN	Contactless	63	61+	48.7	Medium	1





TASK 2.



MERCHANT CATEGORY ANALYSIS

-- Step 1: Create a lookup table for Merchant Category

```
CREATE TABLE Merchant_Category_Mapping
  (Merchant_Category VARCHAR(100) PRIMARY KEY,
   Category_Type VARCHAR(50) NOT NULL);
INSERT INTO Merchant_Category_Mapping (Merchant_Category,Category_Type)
VALUES ('Groceries', 'Essential'),('Online Retail', 'Shopping'),('Healthcare', 'Essential'),
       ('Utilities', 'Essential'),('Gaming', 'Entertainment'),('Gift Cards', 'Shopping'),
       ('Streaming', 'Entertainment'),('Fuel', 'Transportation'),('Electronics', 'Shopping'),
       ('Travel', 'Travel'),('Restaurants', 'Food'),('Crypto Exchange', 'Financial');
SELECT * FROM Merchant_Category_Mapping
```

-- Step 2: Combine transaction data with Merchant Category Mapping

```
SELECT
  c.Transaction_ID,c.Merchant_Category,
  m.Category_Type,c.Transaction_Amt_USD,c.fraud_status
FROM Credit_Card_Fraud c
INNER JOIN
Merchant_Category_Mapping m
on c.Merchant_Category = m.Merchant_Category
```

-- Step 3: Calculate Fraud Rate by Category Type using a CTE

```
;WITH FraudAnalysis as
( SELECT m.category_type, COUNT(*) as Total_Transactions,
  SUM (CASE WHEN c.fraud_status = 1 THEN 1 ELSE 0
    END ) as Fraud_Transactions
FROM Merchant_Category_Mapping m
INNER JOIN Credit_Card_Fraud c
on c.Merchant_Category = m.Merchant_Category
GROUP BY m.Category_Type)
SELECT Category_Type,Total_Transactions,Fraud_Transactions,
CAST( Fraud_Transactions * 100.0 / Total_Transactions
as DECIMAL(5,2)) as Fraud_Rate_Percentage
FROM FraudAnalysis
ORDER BY Fraud_Rate_Percentage DESC
```



Results		Messages			
	Category_Type	Total_Transactions	Fraud_Transactions	Fraud_Rate_Percentage	
1	Financial	561	28	4.99	
2	Entertainment	2046	59	2.88	
3	Shopping	6065	107	1.76	
4	Essential	5809	81	1.39	
5	Transportation	1565	19	1.21	
6	Food	2534	29	1.14	
7	Travel	1420	16	1.13	



TASK 3.

TOP FRAUDULENT TRANSACTION BY MERCHANT CATEGORY

```
;WITH RankedTransactions as
(
  SELECT Transaction_ID,Transaction_Amt_USD,Merchant_Category,
  ROW_NUMBER() OVER (PARTITION BY merchant_category
  ORDER BY transaction_amt_usd DESC) as Transaction_Rank
FROM Credit_Card_Fraud
WHERE Fraud_Status = 1
)

SELECT
  Transaction_ID,
  Transaction_Amt_USD,
  Merchant_Category

FROM RankedTransactions
WHERE Transaction_Rank = 1
ORDER BY Transaction_Amt_USD DESC;
```

Results		Messages	
	Transaction_ID	Transaction_Amt_USD	Merchant_Category
1	13384	3356.46	Gaming
2	10915	3236.34	Crypto Exchange
3	17988	2485.32	Travel
4	4173	1694.19	Electronics
5	17336	1307.2	Groceries
6	1746	597.89	Gift Cards
7	17314	579.84	Utilities
8	13296	527.56	Restaurants
9	15747	359.29	Online Retail



TASK 4.

MERCHANT CATEGORIES WITH HIGH AVERAGE TRANSACTION AMOUNT

```
SELECT
  Merchant_Category,
  COUNT(*) AS Total_Transactions,
  ROUND(AVG(Transaction_Amt_USD), 2) as Average_Transaction_Amount
FROM Credit_Card_Fraud
GROUP BY Merchant_Category
HAVING AVG(Transaction_Amt_USD) > 100
ORDER BY Average_Transaction_Amount DESC;
```

Results Messages			
	Merchant_Category	Total_Transactions	Average_Transaction_Amount
1	Travel	1420	442.41
2	Crypto Exchange	561	357.17
3	Electronics	2032	302.11





TASK 5.

FRAUD RATE BY AUTHENTICATION METHOD

```
SELECT
  Authentication_Method,
  COUNT(*) as Total_Transactions,
  SUM(CASE WHEN Fraud_Status = 1 THEN 1
        ELSE 0
      END) as Fraud_Transactions,
  CAST(SUM(CASE When Fraud_Status = 1 Then 1
            Else 0
          END) * 100.0 / COUNT(*)
        as Decimal (5,2)
    ) as Fraud_Rate_Percentage
FROM Credit_Card_Fraud
GROUP BY Authentication_Method
ORDER BY Fraud_Rate_Percentage DESC;
```

Results		Messages		
	Authentication_Method	Total_Transactions	Fraud_Transactions	Fraud_Rate_Percentage
1	No Authentication	2388	106	4.44
2	OTP	4792	75	1.57
3	3D Secure	5554	86	1.55
4	PIN	4173	48	1.15
5	Biometric	3093	24	0.78





TASK 6.

FRAUD RISK CLASSIFICATION

```
SELECT
    Transaction_ID, Transaction_Amt_USD, Merchant_Category,
    Merchant_Risk_Score, Fraud_Status,
CASE
    WHEN Merchant_Risk_Score >= 80 THEN 'High Risk'
    WHEN Merchant_Risk_Score >= 50 THEN 'Medium Risk'
    ELSE 'Low Risk'
END as Calculated_Risk_Level

FROM Credit_Card_Fraud
WHERE Fraud_Status = 1;
GO
```

Results Messages

	Transaction_ID	Transaction_Amt_USD	Merchant_Category	Merchant_Risk_Score	Fraud_Status	Calculated_Risk_Level
1	64	2396.82	Crypto Exchange	92	1	High Risk
2	149	18.34	Online Retail	18.6	1	Low Risk
3	268	7.25	Gaming	67.1	1	Medium Risk
4	364	246.9	Crypto Exchange	85.7	1	High Risk
5	392	576.76	Crypto Exchange	78.4	1	Medium Risk
6	397	421.53	Groceries	34.6	1	Low Risk
7	404	92.37	Electronics	80.1	1	High Risk
8	600	23	Gift Cards	100	1	High Risk
9	798	28.26	Online Retail	48.7	1	Low Risk





TASK 7.

FRAUD ANALYSIS USING PARAMETERIZED STORED PROCEDURE

```
CREATE PROCEDURE TransactionsByRiskLevel
    @RiskLevel VARCHAR(20)
AS
BEGIN

    SELECT Transaction_ID, Transaction_Amt_USD, Merchant_Category,
        Card_Type, Authentication_Method, Merchant_Risk_Score, Merchant_Risk_Level, Fraud_Status
    FROM Credit_Card_Fraud
    WHERE Merchant_Risk_Level = @RiskLevel
        and Fraud_Status = 1;

END;
GO

-- Execute Procedure
EXEC TransactionsByRiskLevel 'High';
GO
```

Results		Messages						
	Transaction_ID	Transaction_Amt_USD	Merchant_Category	Card_Type	Authentication_Method	Merchant_Risk_Score	Merchant_Risk_Level	Fraud_Status
1	268	7.25	Gaming	Visa	No Authentication	67.1	High	1
2	392	576.76	Crypto Exchange	Visa	OTP	78.4	High	1
3	1556	7.38	Gaming	Visa	3D Secure	66.8	High	1
4	1736	28.98	Crypto Exchange	Mastercard	3D Secure	77.4	High	1
5	2161	48.46	Gift Cards	Mastercard	No Authentication	72.5	High	1
6	2278	50.72	Gift Cards	RuPay	PIN	75.6	High	1
7	2653	164.42	Gaming	Mastercard	No Authentication	65.6	High	1
8	2978	43.69	Gift Cards	Visa	OTP	77.3	High	1
9	3060	5.14	Streaming	RuPay	Biometric	66.8	High	1

